Simulation Arena

Discount Leakage & Cherry-Picking: The Basket Sabotage

Cherry-pick risk 71/100 (Moderate): give this discount and there is a 87.0% chance the Support line or another offsetting line disappears, costing $188k - 38.5% of the deal's profit.

Seed 42 | Run run_a7903e29b106 | 2,000 scenarios tested | Cluster bootstrap resampling with empir… | 21 Sep 2026, 20:10

Same seed, same numbers - this run can be reproduced exactly.

How this works

We went back through your past multi-line deals and found the ones where a line was discounted past 30%. In those deals we measured how often the other lines were cancelled inside 6 months, and compared it with how often those lines die anyway. That difference is the cherry-pick signal. We then rebuilt this quote 2,000 times, each time re-drawing whole past deals so the answer reflects how much history we actually have, and re-priced the basket with whatever survived. The risk score is three things you can check by hand: how far past your discount limit the ask sits (25%), how likely the offsetting lines are to vanish (50%), and how much of the profit sits in the line being cut (25%).

The numbers that matter

CHERRY-PICK RISK SCORE
71 / 100
Built from three checkable pieces; the biggest one is how likely the offsetting lines are to vanish.
vs Moderate risk

PROFIT LIKELY TO LEAK AWAY
$188k
The profit this deal loses once you allow for how buyers actually behave.
vs 38.5% of the deal's profit

CHANCE THE SUPPORT LINE IS DROPPED
64.0%
How often the line that pays for the discount dies in the first 6 months.
vs 2.9x the normal 22.0%

CHANCE ANY OFFSETTING LINE DIES
87.0%
The odds the basket does not survive intact.

PROFIT EXPOSED ON THE SUPPORT LINE
$96k
$150k of profit with a 64.0% chance of walking.

PROFIT AT RISK
38.5%
Share of the bundle's profit that is exposed to cherry-picking.

PROFIT IF THEY KEEP EVERYTHING
$489k
The spreadsheet number: 54.3% margin on $900k.

PROFIT YOU SHOULD ACTUALLY PLAN FOR
$301k
The honest number after cherry-picking.
vs as low as $0 in a bad run

CHANCE THE BUNDLE ENDS UNDER YOUR 35% FLOOR
14.2%
How often the surviving basket is not worth signing at all.

How we got there

1. What is actually on the table

The basket is worth $900k and makes $489k if nothing is dropped. But 61.3% of that profit sits in Software - the exact line the buyer wants 35% off, which is 5 points past the discount line you set yourself.

Evidence: $300k of $489k profit in Software

2. What history says happens next

You gave us no history to learn from, so we used your own numbers: 22.0% of lines die anyway, and 64.0% die once a deep discount is given.

Evidence: source: the rates you typed in

3. What the rebuild shows

Rebuilding this quote 2,000 times, the profit lands at $301k on average, and in an unlucky run of twenty it falls to $0. There is a 14.2% chance the bundle that survives is worth less than your 35% margin floor.

Evidence: $188k average shortfall

4. What to do about it

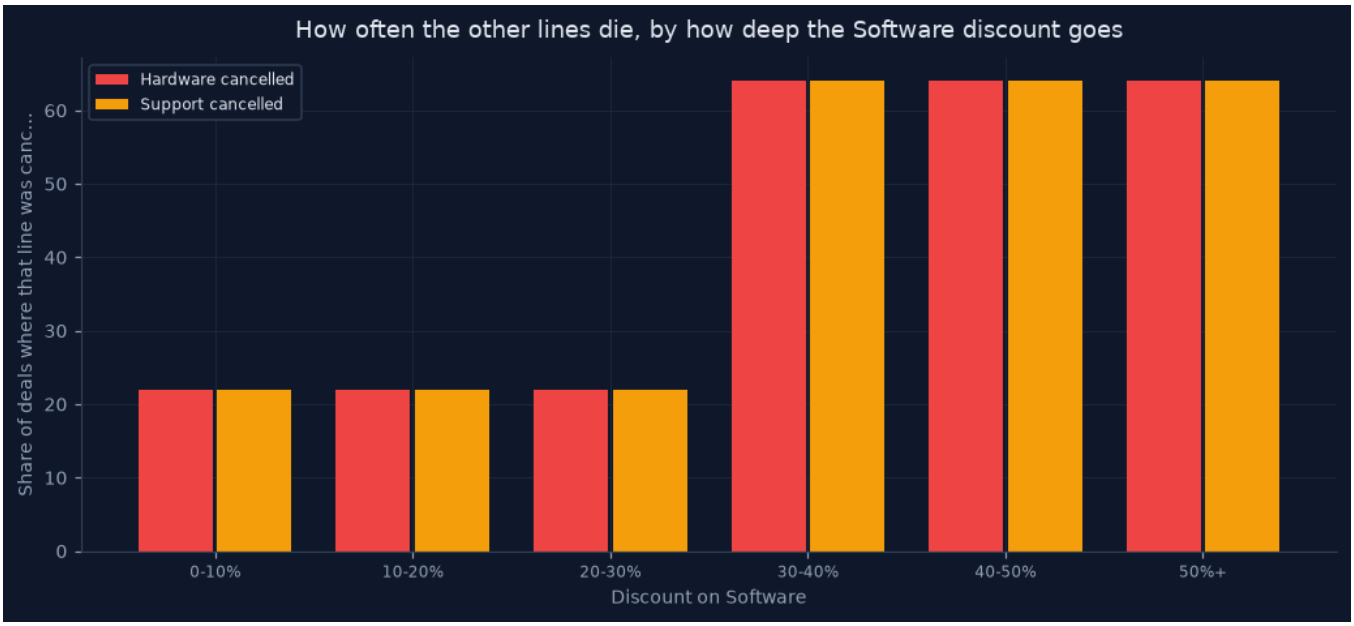
The biggest driver of the score is how likely the offsetting lines are to vanish. Tie the discount to the behaviour: if the Support line is not live at 80% of committed hours by month 6, the 35% Software discount comes off and Software reverts to standard pricing - worth up to $210k back.

Evidence: clawback up to $210k

Charts

How often the other lines die, by how deep the Software discount goes

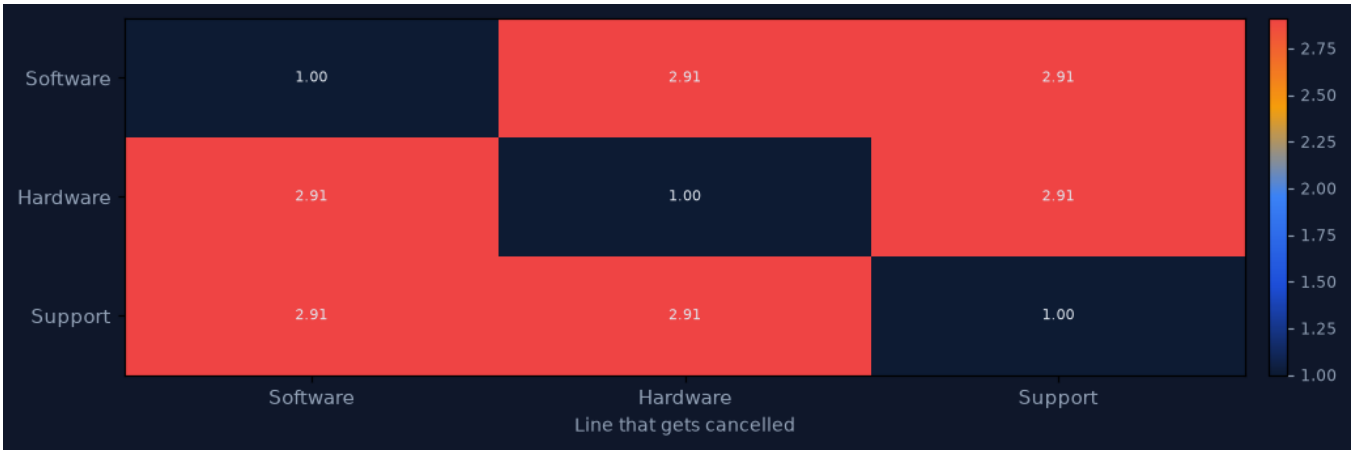
No history supplied - this is your assumption, drawn as a step



What this shows: If the bars jump once the discount passes your limit, buyers are not just buying more - they are buying differently. That jump is the cherry-pick signal.

Which discount poisons which line

Numbers are 'times more likely than usual' when the row's line is discounted past 30%



What this shows: Anything above 1.0 means the two events travel together. The hottest cell is the pair you should write a contract clause about.

What this deal actually earns

2,000 rebuilt bundles - the dashed line is the number on the quote



What this shows: The gap between the tall bars and the 'if nothing is dropped' marker is the cost of cherry-picking. The left tail is the bad case you are signing up for.

Tables

The quote, line by line

Where the profit sits, and how exposed each line is.

Line	List price	Discount	Price after discount	Your margin	Profit	Share of profit	Chance it is dropped	Profit at risk
Software	$600,000	35.0%	$390,000	85.0%	$300,000	61.3%	22.0%	$66,000
Hardware	$300,000	5.0%	$285,000	18.0%	$39,000	8.0%	64.0%	$24,960
Support	$250,000	10.0%	$225,000	70.0%	$150,000	30.7%	64.0%	$96,000

Row 1 carries the discount being asked for.

What happens to each line when another line is discounted past 30%

Read across a row: 'when THIS line is deeply discounted, the chance THAT line dies'. A figure above 1.0 means the two events travel together.

Discounted line (A)	Line that dies (B)	Deals like this	Normal rate	Rate after the discount	Times more often	Link strength
Software	Software	-	22.0%	22.0%	1.00x	-
Software	Hardware	-	22.0%	64.0%	2.91x	assumed
Software	Support	-	22.0%	64.0%	2.91x	assumed
Hardware	Software	-	22.0%	64.0%	2.91x	assumed
Hardware	Hardware	-	22.0%	22.0%	1.00x	-
Hardware	Support	-	22.0%	64.0%	2.91x	assumed
Support	Software	-	22.0%	64.0%	2.91x	assumed
Support	Hardware	-	22.0%	64.0%	2.91x	assumed
Support	Support	-	22.0%	22.0%	1.00x	-

No history was available - these are your assumptions, not evidence.

What the bundle actually earns

Every one of the 2,000 rebuilt bundles, summarised.

Scenario	Profit	Margin	Versus the plan
If they keep everything	$489,000	54.3%	the number on the quote
Most likely outcome	$300,537	33.4%	-$188,463
A bad run (1 in 20)	$0	0.0%	-$489,000
Worst bundle we saw	$0	0.0%	-$489,000

Row 2 is the number to plan with.

What we recommend

Approve the 35% only as a conditional discount - expect to lose $188k if you do not tie it to the Support line.

Risk Moderate 71 out of 100 | Confidence: Low

Why

• 61.3% of this deal's profit sits in Software, the line being cut by 35%.

• Past deals like this one lose the offsetting lines 87.0% of the time (the rates you typed in).

• Expected profit after cherry-picking is $301k, not $489k - a $188k gap.

What to do

1. Make the 35% Software discount conditional: if the Support line is not live at 80% of committed hours by month 6, Software pricing reverts to the standard tier (recovering up to $210k).

2. Do not let them drop one offsetting line and keep the other: make the discount conditional on the whole Hardware, Support basket surviving, not on any single line.

3. Put the test at month 6, not at signature.

4. There is a 14.2% chance the surviving bundle sits under your 35% floor - require CF sign-off above 30% on any single line.

5. Upload past quote lines with discount and cancellation flags so this number comes from evidence instead of your assumption.

The trade-off: Walking away from the 35% risks the deal; giving it unprotected costs about $188k. The clause lets you keep the deal and cap the loss. If the buyer refuses any condition, the mildest version is to cap Software at 30%, which still protects $30k.

What we assumed and what to watch

Worth knowing before you act

• No usable history was mapped (we need a category, the discount given and whether the line was cancelled), so every probability here is your typed-in assumption: 22.0% normal abandonment, 64.0% after a deep discount.

Assumptions

• Quote under review: Software $600,000 at 35% off (85.0% margin), Hardware $300,000 at 5% off (18.0% margin), Support $250,000 at 10% off (70.0% margin).

• Deep discount means anything past 30%; a line counts as abandoned if it dies within 6 months.

• Abandonment rates come from the rates you typed in: 22.0% normal, 64.0% after a deep discount.

• Risk score = 25% x how far past your discount limit the ask sits (10 points past = full marks) + 50% x chance an offsetting line dies + 25% x share of profit sitting in the discounted line.

• A cancelled line is treated as fully lost: no revenue and no cost.

• Bundle margin floor 35%; minimum support commitment 80% of hours.

• We replayed 2,000 bundles built from your own past deals; seed 42 - re-running gives identical numbers.

- Quote / deal ID -> not in your file; not used
- Customer tier / segment -> not in your file; not used
- Line item ID -> not in your file; not used
- Product line name -> not in your file; not used
- Line category -> not in your file; not used
- List price of the line -> not in your file; not used
- Cost of the line -> not in your file; not used
- Discount on the line (%) -> not in your file; not used
- Was the line cancelled? (1/0) -> not in your file; not used
- Months until cancellation -> not in your file; not used

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
0.00s	prepare	Line category -> not in your file; not used
0.00s	prepare	List price of the line -> not in your file; not used
0.00s	prepare	Cost of the line -> not in your file; not used
0.00s	prepare	Discount on the line (%) -> not in your file; not used
0.00s	prepare	Was the line cancelled? (1/0) -> not in your file; not used
0.00s	prepare	Months until cancellation -> not in your file; not used
0.00s	prepare	Checking your assumptions…
0.00s	simulate	Running the simulation…
0.00s	read	Reading the quote you want us to stress-test...
0.00s	read	The basket is worth $900k and carries $489k of profit (54.3% margin) if the buyer keeps every line.
0.00s	read	The line being discounted hardest is Software at 35% off - that is 5 points past your 30% deep-discount line.
0.00s	read	No past quote lines were supplied, so we will use the rates you typed in instead of learning from evidence.
0.00s	prepare	Checking what happened in past deals that got a discount this deep...
0.00s	prepare	Chance the offsetting lines disappear: 87.0% - that is what the discount is really buying.
0.00s	simulate	Rebuilding 2,000 bundles that look like this one and re-pricing each of them...
0.00s	simulate	Rebuilt 10% of the bundles...
0.00s	simulate	Rebuilt 20% of the bundles...
0.00s	simulate	Rebuilt 30% of the bundles...
0.00s	simulate	Rebuilt 40% of the bundles...
0.00s	simulate	Rebuilt 60% of the bundles...
0.00s	simulate	Rebuilt 70% of the bundles...
0.00s	simulate	Rebuilt 80% of the bundles...
0.00s	simulate	Rebuilt 90% of the bundles...
0.00s	summarize	All bundles priced. Working out what this deal really earns...
0.00s	summarize	Done. Cherry-pick risk 71/100 (Moderate) - expect to lose $188k unless the discount is made conditional.